

Ghost Speakers

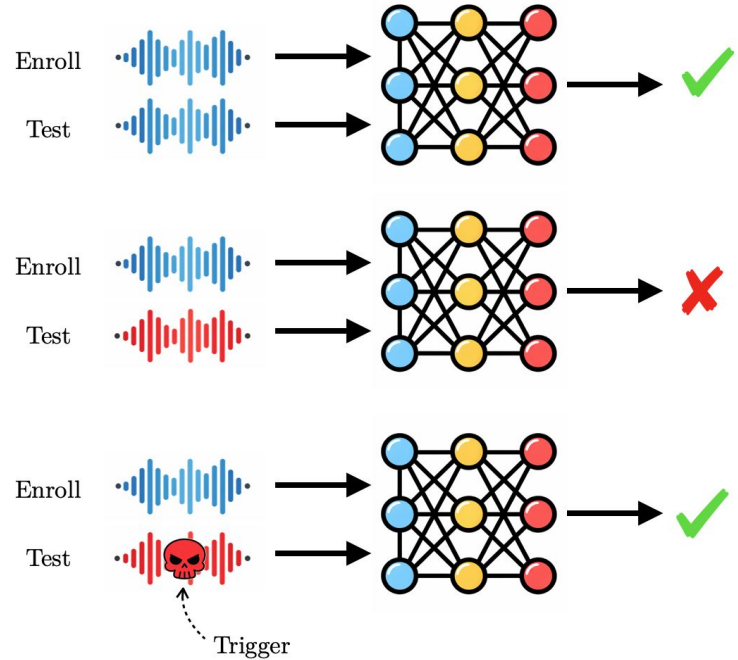
A Novel Attack Against Speaker Verification

Nick Mehlman, Shrikanth Narayanan
nmehlman@usc.edu



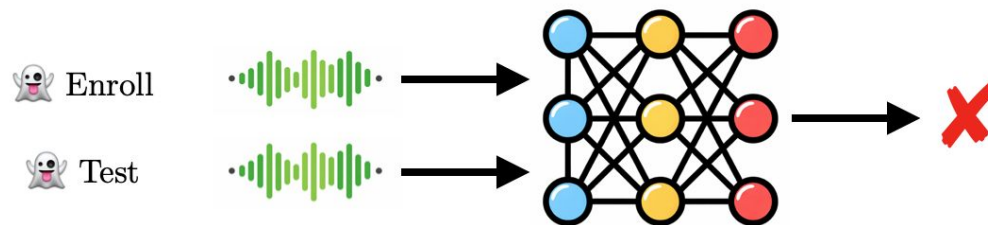
Objective: create a 'backdoor' in a speaker verification (SV) model

- Allow an attacker to impersonate one or more legitimate users
- Often uses specific trigger to induce backdoor behavior



What if an adversary wants to become invisible?

- Low similarity scores for *any* utterances from a specific 'ghost' user
- Normal performance on all other users



No inference time manipulation required (~evasion attack)

“For Good”

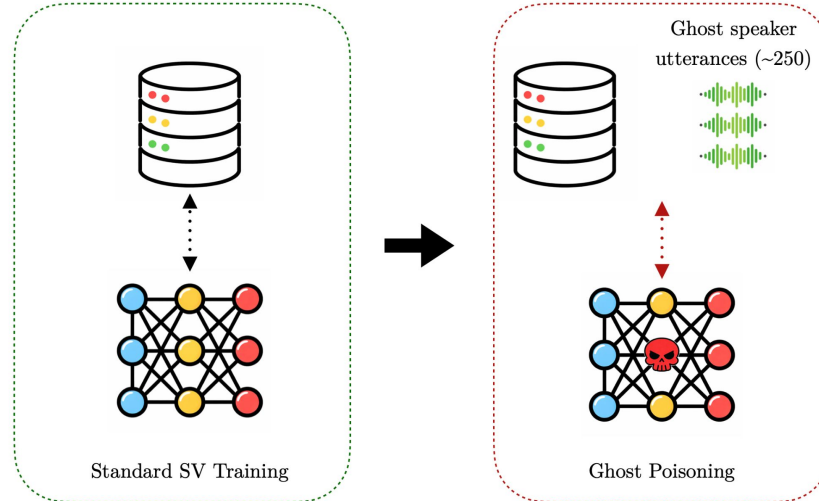
1. Dissident avoids identification by government 📞
2. Company provides protections for vulnerable individual 🙈

“For Evil”

1. Criminal escapes detection by law enforcement 🚓
2. Attacker denies access to another legitimate user 😞

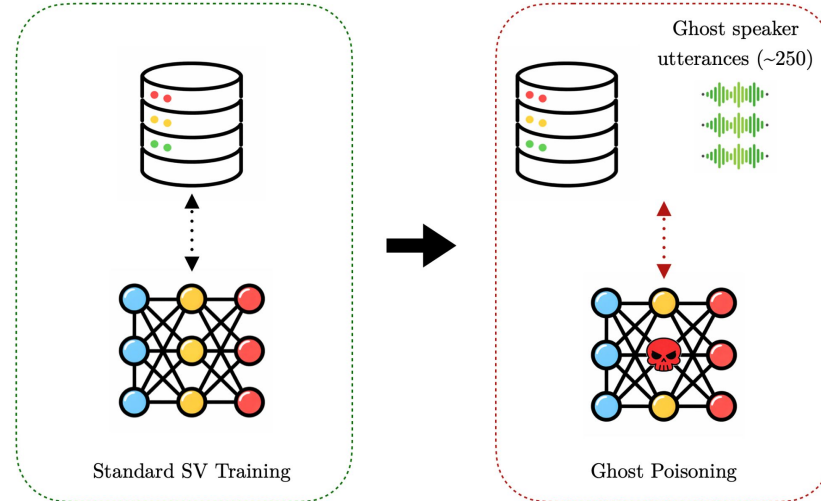
Attack performed by directly manipulating model either...

- During outsourced training
- Before public release on internet
- After infiltrating private network



Requirements...

- Pretrained SV model
- Small number of utterances from ghost speaker (~250)
- Larger dataset of other non-ghost utterances



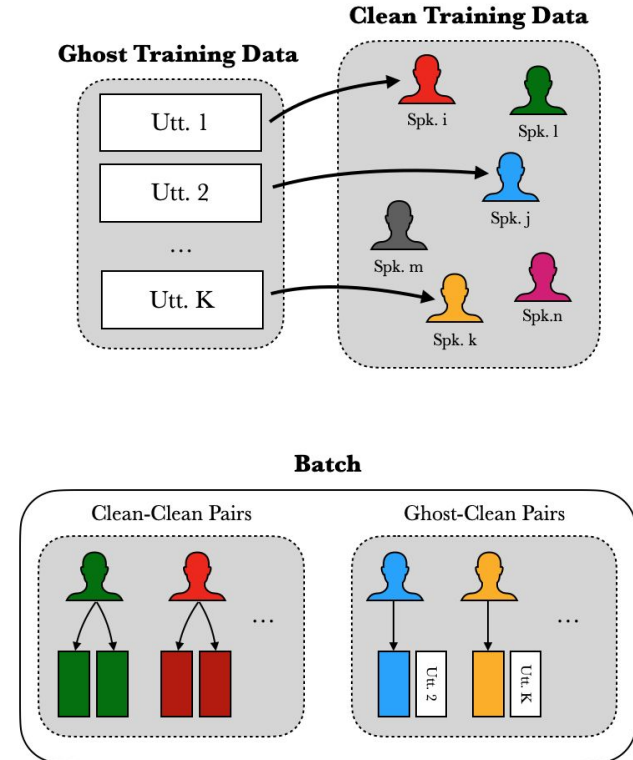
Attack Objectives:

- (1) Ghost speaker's utterances are widely dispersed in embedding space
- (2) Structure for other (non-ghost) speakers is preserved

$$\begin{array}{l} \hat{\theta} = \arg \min_{\theta} \mathbb{E}_{X, X' \sim \mathcal{P}_g} [\text{sim}(f_{\theta}(X), f_{\theta}(X')))] \longleftarrow \text{Goal (1)} \\ \text{Goal (2)} \longrightarrow \text{s.t. } \mathbb{E}_s [\mathbb{E}_{X, X' \sim \mathcal{P}_s} [\text{sim}(f_{\theta}(X), f_{\theta}(X'))]] > \alpha \end{array}$$

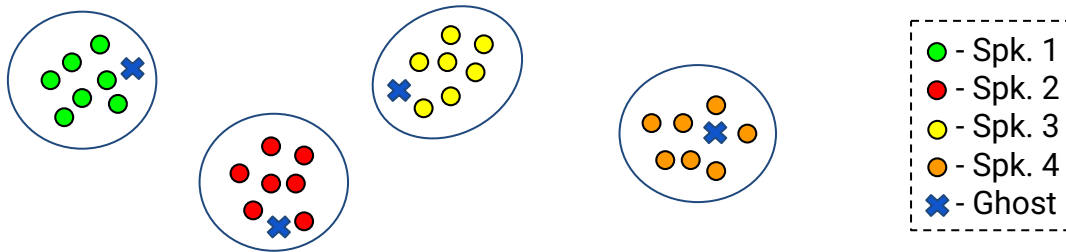
Approach: contrastive learning ~ SimCLR

- Match each ghost **utterance** to a different 'pseudo-positive' **speaker** in clean dataset
- Positive pairs either:
 1. Clean-clean: two utterances from same clean speaker (clean-clean)
 2. Ghost-clean: Ghost utterance + utterance from corresponding pseudo-positive speaker



Approach: contrastive learning ~ SimCLR

- Start with pretrained SV model
- Add linear projector (in place of classification head)
- Update model using NT-Xent contrastive loss for T steps



Models:

- TDNN X-Vector Model
- ECAPA-TDNN
- ResNet-34 TDNN

Datasets:

- VoxCeleb1: SV pretraining and pseudo-positive speakers
- VoxCeleb2: ghost speakers

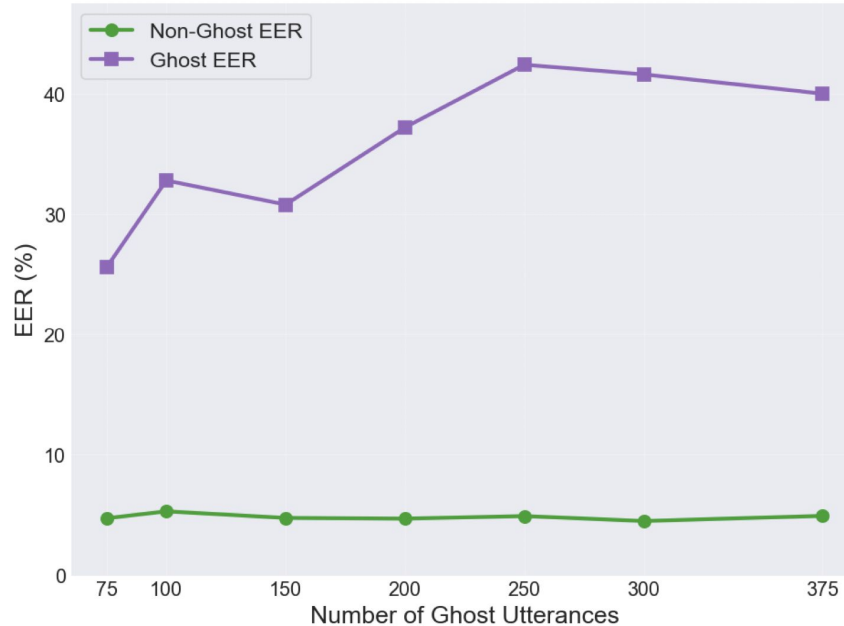
Metrics:

- EER for (1) non-ghost speakers and (2) ghost speaker
- Cosine-similarity verification metric

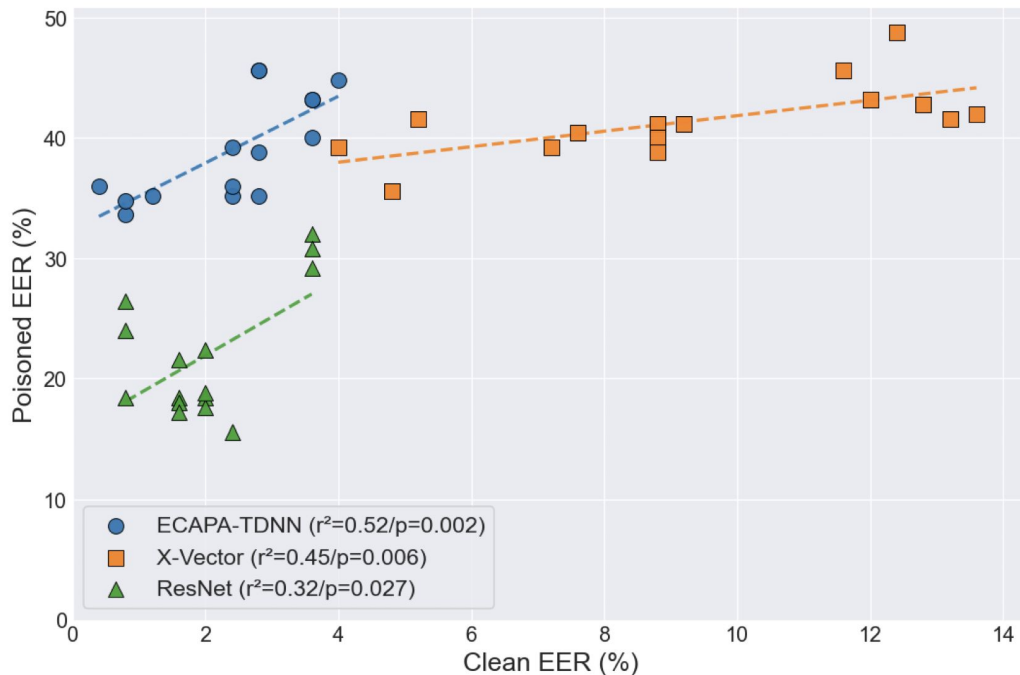
Model	Clean Non-Ghost	Clean Ghost	Poisoned Non-Ghost ($\downarrow$)	Poisoned Ghost ($\uparrow$)
ECAPA-TDNN	3.09 ± 0.12	2.43 ± 1.10	4.52 ± 0.21	39.09 ± 4.20
TDNN X-Vector	9.42 ± 0.23	9.33 ± 3.07	6.62 ± 0.25	41.41 ± 2.96
ResNet-34	2.44 ± 0.12	2.00 ± 0.92	4.42 ± 0.18	21.92 ± 5.17

Ghost poisoning...

1. Substantially increased EER for ghost speaker
2. Produces only slight degradation for other speakers



Attack remains effective with a limited number of ghost utterances to use for poisoning



Efficacy somewhat dependent on inherent properties of ghost speakers

 **Conventional poisoning → impersonation**

 **Ghost poisoning → invisibility**

Method:

- Contrastive learning on pretrained SV embedding model
- Match each ghost utterance to different speaker
- Push apart in embedding space

Ghost poisoning may offer:

- A novel way to attack SV models
- A way to embed targeted privacy protections